

Efficiently Identifying Task Grouping for Multi-Task Learning

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1. Multi-Task Learning

2. TAG

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1. Multi-Task Learning

Prior Works

- Multi-Task Learning은 여러 Task를 수행하는 모델을 학습시키는 방법론
- Three ways to approach multi-task learning

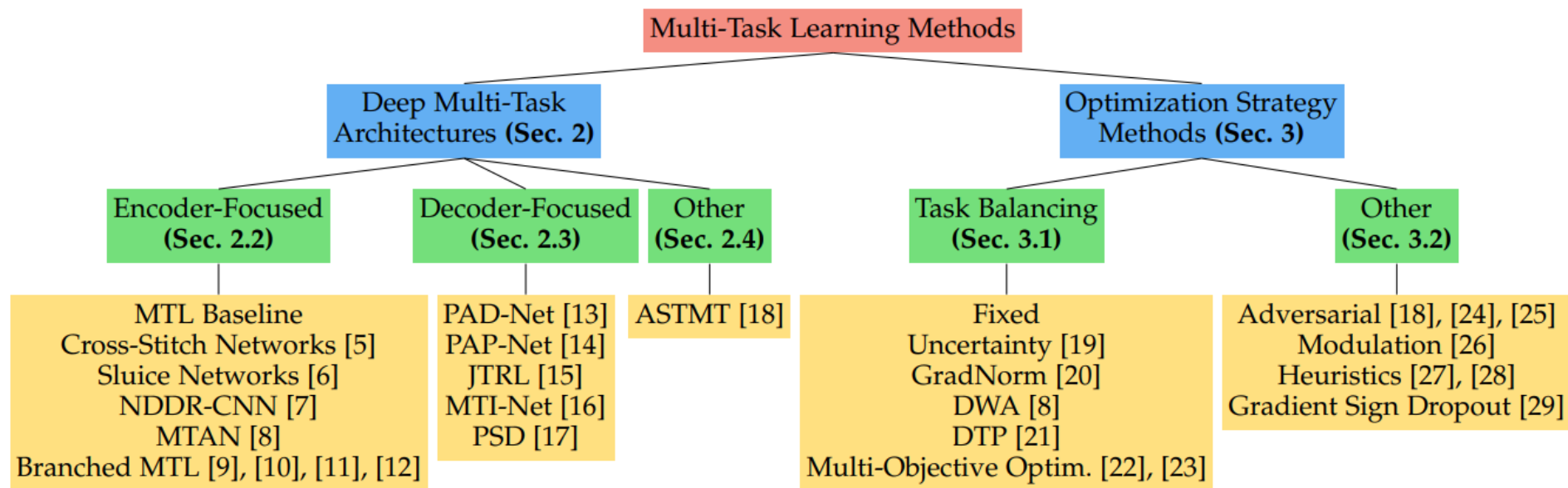


Fig1. A taxonomy of deep learning approaches for jointly solving multiple dense prediction tasks



Prior Works

- Three ways to approach multi-task learning

1. Architectural approach

- How to configure the architecture to perform multi-task learning
- Hard** Parameter Sharing / **Soft** Parameter Sharing
- Encoder-focused model / Decoder-focused model

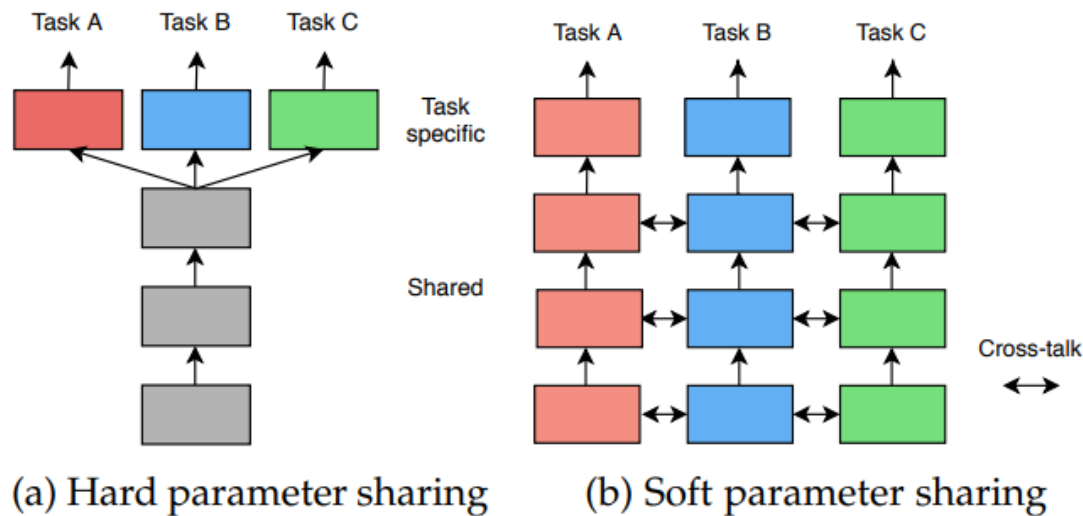


Fig2. Hard / Soft parameter sharing schemes

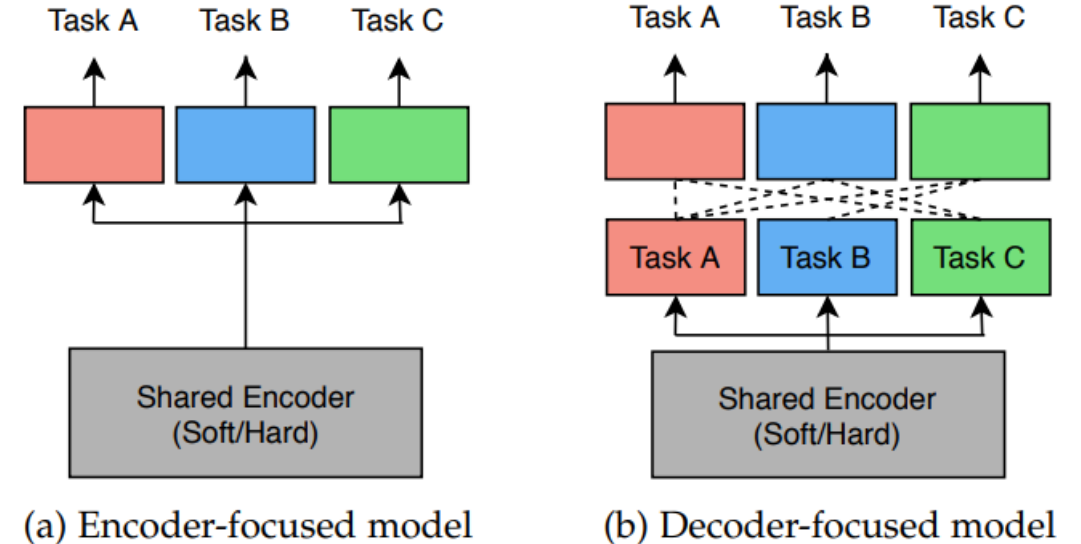


Fig3. Encoder / Decoder-focused models



Prior Works

2. Task Grouping

- Perform Multi-Task Learning considering Relevance between Tasks

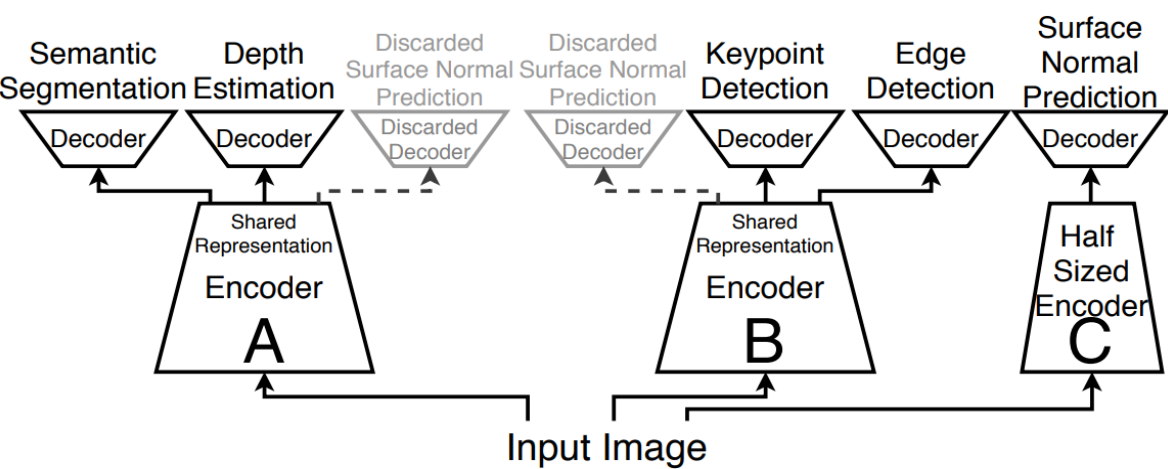


Fig4. Task Grouping^[1]

		Relative Performance On					Average
		SemSeg	Depth	Normals	Keypoints	Edges	
Trained With	SemSeg	—	-5.41%	-11.29%	-4.32%	-34.64%	-13.92%
	Depth	4.17%	—	-3.55%	3.49%	3.76%	1.97%
	Normals	8.50%	2.48%	—	1.37%	12.33%	6.17%
	Keypoints	4.82%	1.38%	-0.02%	—	-5.26%	0.23%
	Edges	3.07%	-0.92%	-4.42%	1.37%	—	-0.23%
	Average	5.14%	-0.62%	-4.82%	0.48%	-5.95%	-1.15%

Fig5. Pair-wise multi-task relationship

[1] Which Tasks Should Be Learned Together in Multi-task Learning? (2020 ICLR)



Prior Works

3. Loss Weight / Gradient

- Learning speed and Loss Landscape of each task is different
- Loss weights and/or gradient adjustment

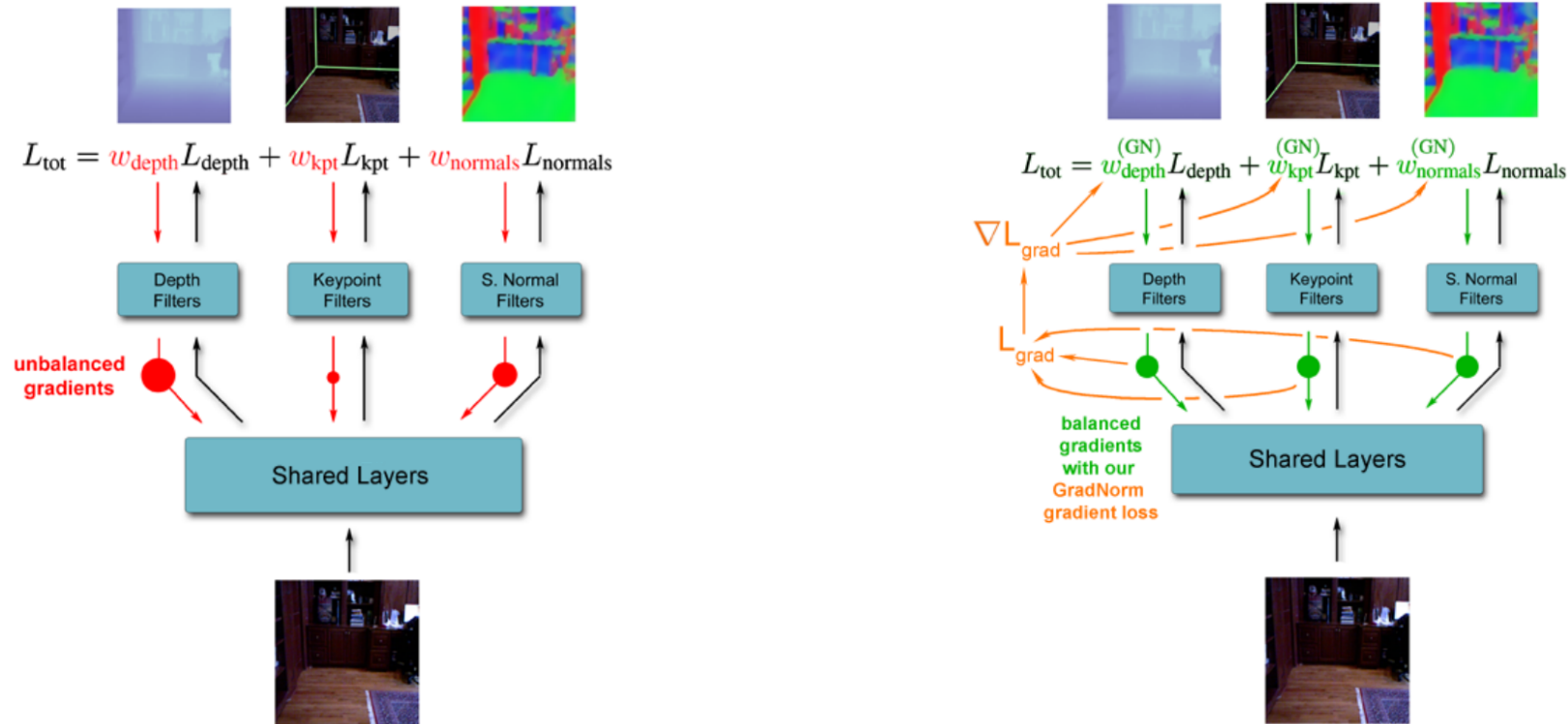


Fig6. Without GradNorm and with GradNorm



Question

2. *TAG*



Motivation

- Tends to rely on intuition and experience
- ***To further complicate matters, systematic technique to group the task is needed***
 - ✓ Naïve Method : exhaustively search for $2^{|n|} - 1$ multi-task networks

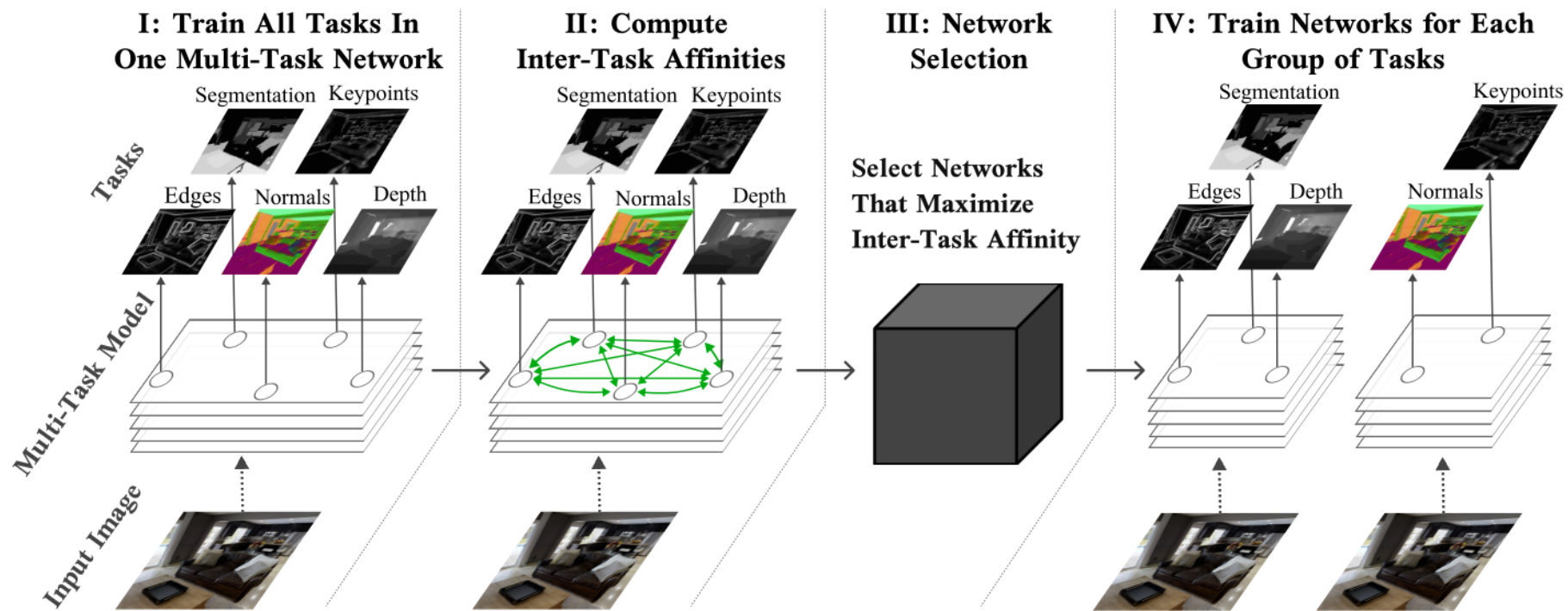


Fig7. Suggeste task grouping method



Problem Definition

- Given a set of tasks T , a fixed inference-time memory budget b , and latency constraints c ,
- Assign tasks to networks that average task performance is maximized
 - each network must have parameter count less than c (?)
 - ✓ Networks must span our set of tasks τ
 - ✓ Total number of networks is less than or equal to our memory budget b
- Let's formalize above circumstances,
 - ✓ A set of n tasks $T = \{\tau_1, \tau_2, \dots, \tau_n\}$
 - ✓ k multi-task neural networks $M = \{m_1, m_2, \dots, m_k\}$, $k \leq b$
 - ✓ For a given performance measure P , aggregate performance of task grouping is as follows:

$$\sum_{i=1}^n P(\tau_i | M)$$

where $P(\tau_i | M)$ computes the performance of task τ_i from the set of models M using the model m_j which the task grouping algorithm predicts the performance of τ_i will be highest



Inter-Task Affinity

- By sharing parameters, tasks implicitly transfer information to each other
- This paper propose to view the task's successive gradient updates affect the loss of other tasks
- Total Loss can be defined as

$$L_{total}(\mathcal{X}, \theta_s \{\theta_i\}) = \sum_{i \in T} L_i(\mathcal{X}, \theta_s, \theta_i)$$

for simplicity of notation, loss weight of each task is set to 1

L : Loss
$\mathcal{X}$: training batch
θ_s : shared parameter
θ_i : task-specific parameter

- For a given training batch $\mathcal{X}^t$ at time-step t , quantity $\theta_{s|i}^{t+1}$ represent the updated shared parameter
- If stochastic gradient descent is used, updated shared parameter with respect to task i is written as follows:

$$\theta_{s|i}^{t+1} := \theta_s^t - \eta \nabla_{\theta_s^t} L_i(\mathcal{X}^t, \theta_s^t, \theta_i^t)$$

- Now, we can *lookahead* loss for each task by using the updated shared parameters



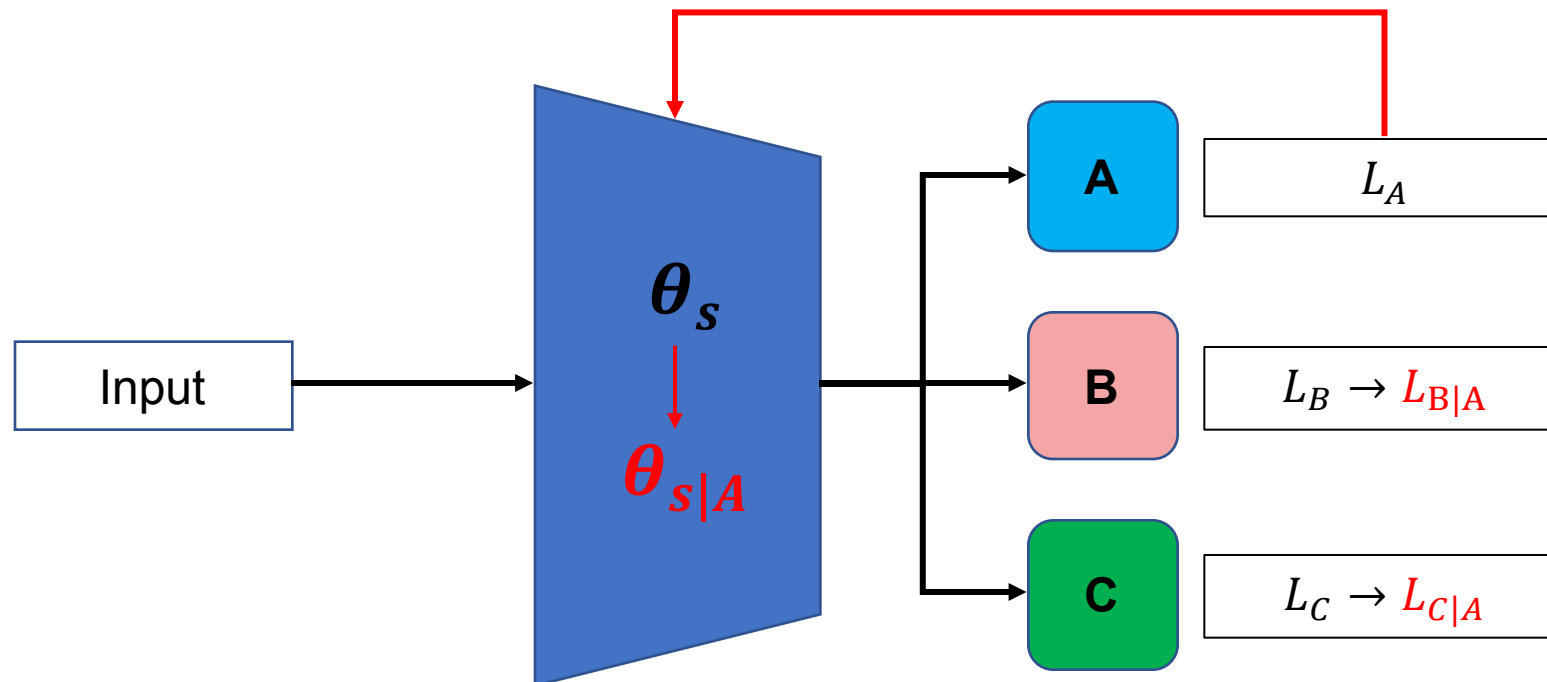
Inter-Task Affinity

- To assess the effect of the gradient update of task i on a given task j , compare the loss of task j before and after

$$Z_{i \rightarrow j}^t = 1 - \frac{L_j(\mathcal{X}^t, \theta_{s|i}^{t+1}, \theta_j^t)}{L_j(\mathcal{X}^t, \theta_s^t, \theta_j^t)}$$

Z : affinity

To eliminate the scale discrepancy among different tasks, ration of task's loss is used



$$Z_{A \rightarrow B} = 1 - \frac{L_{B|A}}{L_B}$$
$$Z_{A \rightarrow C} = 1 - \frac{L_{C|A}}{L_C}$$



Task Affinity Grouping (TAG)

- After training all tasks together in one model, pairwise task affinities are calculated
 - Formally, $\binom{|\tau|}{2}$ values are collected during a single training
- Among $2^{|\tau|} - 1$ possible networks, k multi-task networks are selected under inference-time memory budget limitation b
- Also, every task must be served from exactly one network inference (Set-Cover problem)
- For task groupings consist of three or more tasks, TAG approximates inter-task affinity by leveraging pairwise affinities

$\{A\},\{B\},\{C\}$

Tasks	A	B	C
A	$Z_{A \rightarrow A}$	$Z_{A \rightarrow B}$	$Z_{A \rightarrow C}$
B	$Z_{B \rightarrow A}$	$Z_{B \rightarrow B}$	$Z_{B \rightarrow C}$
C	$Z_{C \rightarrow A}$	$Z_{C \rightarrow B}$	$Z_{C \rightarrow C}$

$\{A,B\},\{C\}$

Tasks	A	B	C
A	$Z_{A \rightarrow A}$	$Z_{A \rightarrow B}$	$Z_{A \rightarrow C}$
B	$Z_{B \rightarrow A}$	$Z_{B \rightarrow B}$	$Z_{B \rightarrow C}$
C	$Z_{C \rightarrow A}$	$Z_{C \rightarrow B}$	$Z_{C \rightarrow C}$

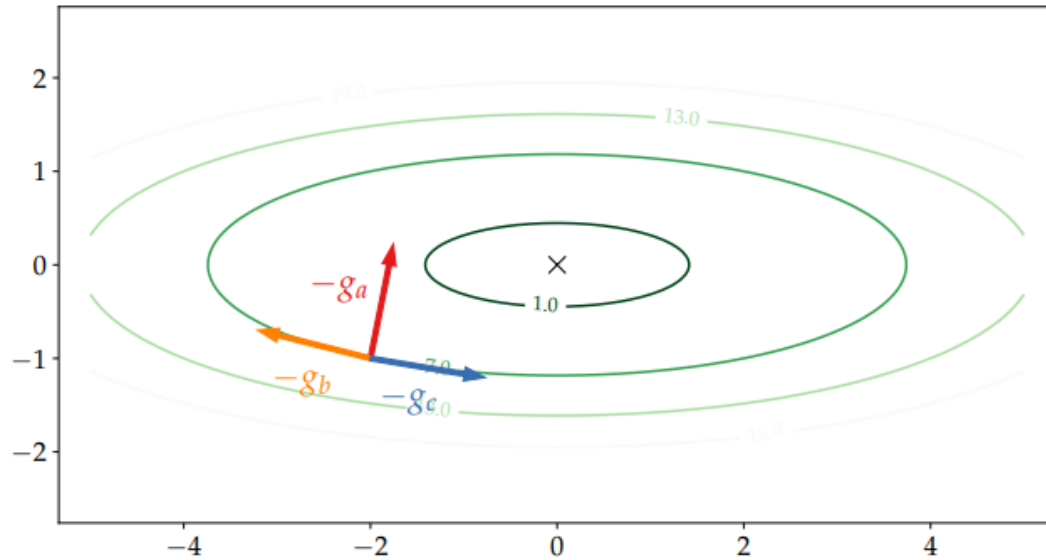
$\{A,B,C\}$

Tasks	A	B	C
A	$Z_{A \rightarrow A}$	$Z_{A \rightarrow B}$	$Z_{A \rightarrow C}$
B	$Z_{B \rightarrow A}$	$Z_{B \rightarrow B}$	$Z_{B \rightarrow C}$
C	$Z_{C \rightarrow A}$	$Z_{C \rightarrow B}$	$Z_{C \rightarrow C}$

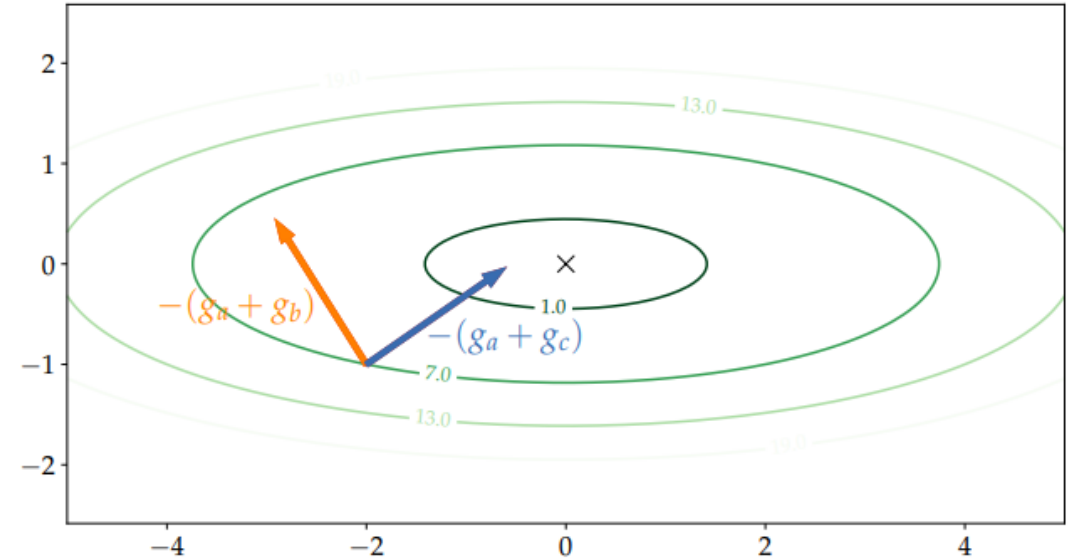


Counter Example

- Let's assume that task **B** induces higher affinity than **C** on **A**
- Then, according to TAG, grouping $\{A,B\}$ is done rather than $\{A,C\}$
- Really?



(a) The gradient g_b reduces the loss $\mathcal{L}_a$ more so than g_c .



(b) The combined gradient $g_a + g_c$ reduces the loss $\mathcal{L}_a$ more so than $g_a + g_b$.



Theoretical Analysis

- Two assumption
 - Suppose that in a given step, task B has higher inter-task affinity than task C on task A
 - Suppose that gradients have equal norm, i.e. $\|g_a\| = \|g_b\| = \|g_c\|$
- If we show the inequality below, TAG can be said to be valid:

$$L_a(\theta - \eta(g_a + g_b)) \leq L_a(\theta - \eta(g_a + g_c))$$

- If **upper-bound** of left side $\leq$ **lower-bound** of right-side, TAG is valid
- Let f be β -**strongly smooth** loss function,

$$|f(x) - f(y) - \nabla f(y)(x - y)| \leq \frac{\beta}{2} \|x - y\|^2$$



Theoretical Analysis

- By substituting L_A for f , y for θ , x for $\theta - \eta(g_a + g_b)$, we get the upper bound of $L_a(\theta - \eta(g_a + g_b))$

$$\begin{aligned} L_a(\theta - \eta(g_a + g_b)) &\leq L_a(\theta) - \eta g_a \cdot (g_a + g_b) + \frac{\eta^2 \beta}{2} \|g_a + g_b\|^2 \\ &= L_a(\theta) + (\eta^2 \beta - \eta) g_a \cdot g_b + \left(\frac{\eta^2 \beta}{2} - \eta\right) \|g_a\|^2 + \frac{\eta^2 \beta}{2} \|g_b\|^2 \end{aligned}$$

- Similarly, we get the lower bound of $L_a(\theta - \eta(g_a + g_c))$

$$L_a(a) - \eta g_a (a_a + g_c) + \frac{\eta^2 \alpha}{2} \|g_a + g_c\|^2 \leq L_a(\theta - \eta(g_a + g_c))$$

- By making another inequality with bounds of each side, we can get

$$(\eta\beta - 1) g_a \cdot g_b + \frac{\eta\beta}{2} (\|g_a\|^2 + \|g_b\|^2) \leq (\eta\alpha - 1) g_a \cdot g_c + \frac{\eta\alpha}{2} (\|g_a\|^2 + \|g_c\|^2)$$



Theoretical Analysis

- According to the Lemma 1 in the paper,

$$g_a \cdot g_c - \frac{\beta \eta}{2} \|g_c\|^2 + \frac{\alpha \eta}{2} \|g_b\|^2 \leq g_a \cdot g_b$$

- Therefore, the $g_a \cdot g_b$ part of the inequality can be substituted by using Lemma above,

$$\begin{aligned} (\eta\beta - 1) \left(g_a \cdot g_c + \frac{\eta\alpha}{2} (\|g_b\|^2) - \frac{\eta\beta}{2} (\|g_c\|^2) \right) + \frac{\eta\beta}{2} (\|g_a\|^2 + \|g_b\|^2) \\ \leq (\eta\alpha - 1) g_a \cdot g_c + \frac{\eta\alpha}{2} (\|g_a\|^2 + \|g_c\|^2) \end{aligned}$$

- After rearranging the inequality above, we can get

$$0 \leq \|g_a\|^2 \left(\frac{\eta}{2} (\alpha - \beta) \right) + \|g_b\|^2 \left(-\frac{\eta}{2} \alpha (\eta\beta - 1) \right) + \|g_c\|^2 \left(\frac{\eta}{2} (\alpha - \beta) + \frac{\eta^2}{2} \beta^2 \right) + \eta(\alpha - \beta) g_a \cdot g_c$$

- Under mild assumption $\|g_a\| = \|g_b\| = \|g_c\|$,

$$\cos(g_a, g_c) \leq \frac{\eta}{4} \frac{\alpha\beta}{\beta - \alpha} - 1$$



Question

3. Experiments



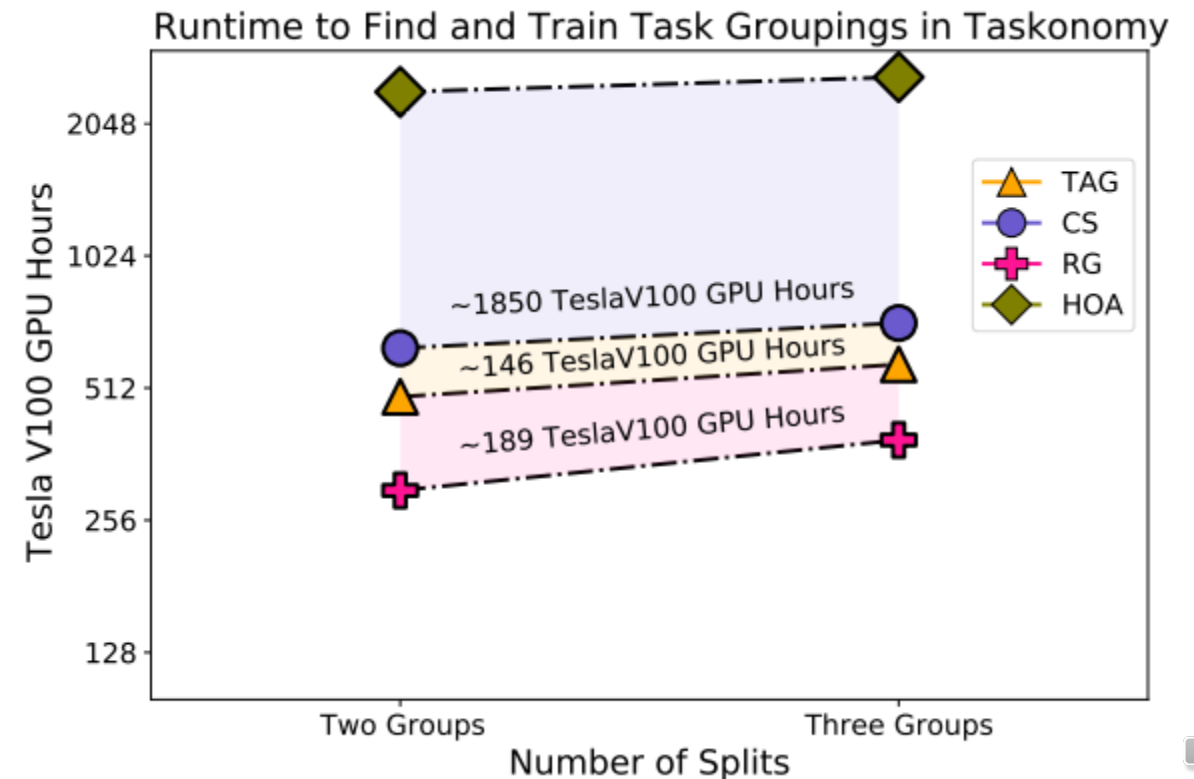
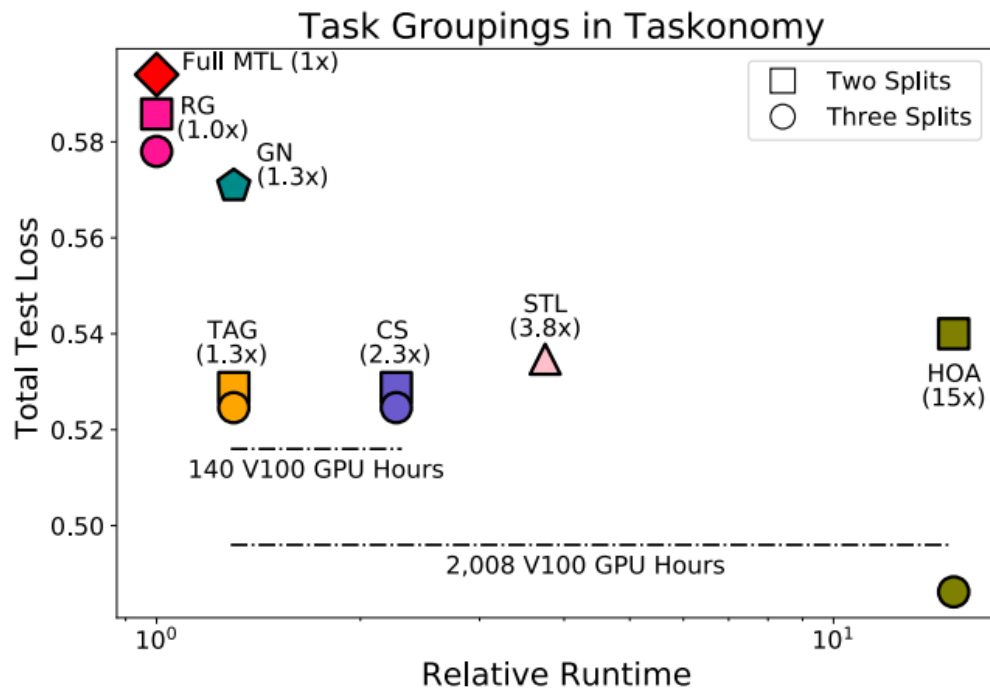
Experiments settings

- Questions
 - 1) Does our measure of inter-task affinity align with identifying which tasks should train together?
 - 2) How do inter-task affinities change over the course of training?
- Experiments
 - I. Effect of Task grouping
 - ✓ MTL : train all task together
 - ✓ STL : train every task to itself
 - ✓ RG : randomly select tasks
 - ✓ CS : inter-task cosine similarity
 - ✓ TAG : paper's method
 - ✓ HOA : another task-grouping method
 - II. Effect of optimization
 - ✓ Uncertainty Weights (UW)
 - ✓ GradNorm (GN)
 - ✓ PCGrad



Time to determine task groupings

- Time to determine task groupings (Left Figure)
 - ✓ TAG has lowest test Error / Loss
 - ✓ TAG only gets slower about x1.3 than RG to make task groups
- Time to determine task groupings and train task groups (Right Figure)



Multi-task Ablation Study

- Q1. Does our measure of inter-task affinity align with identifying which tasks should train together?
 - ✓ Performance found by TAG correlates well to optimal
 - ✓ Exception occurred in a8
 - ✓ Affinity score difference between best and worst task is just 0.04

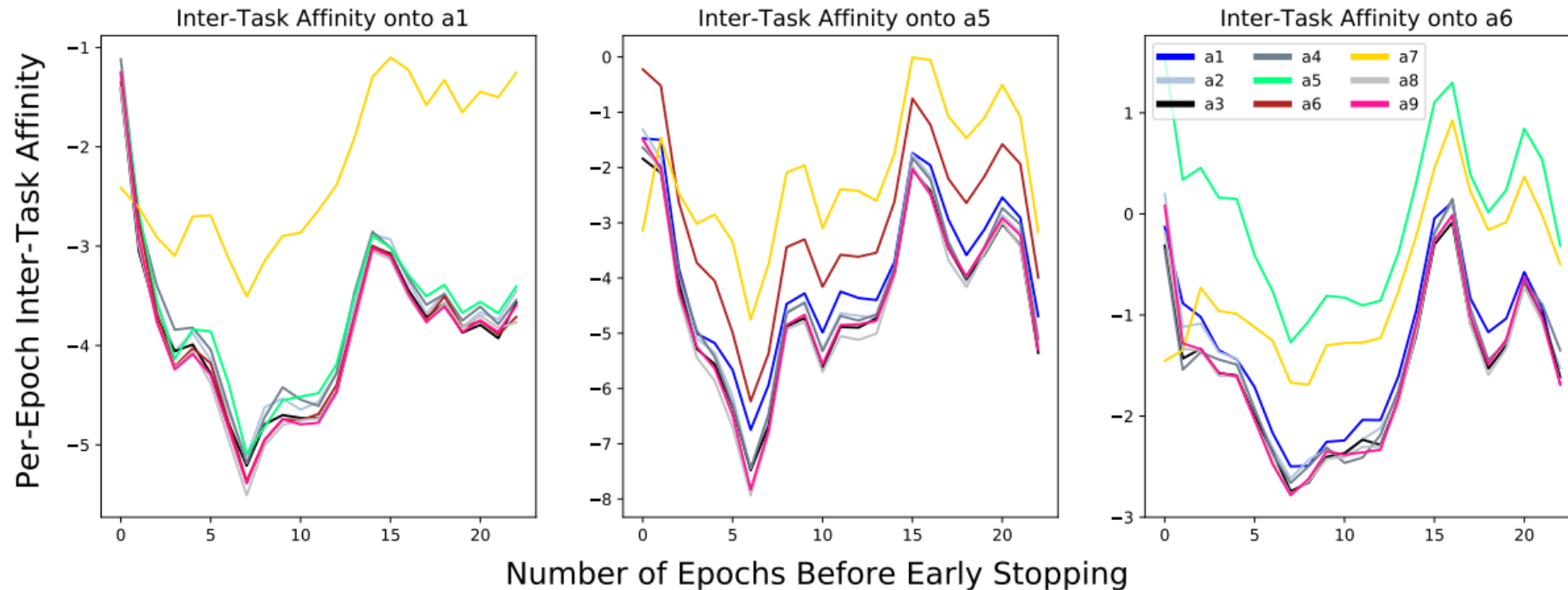
Tasks	Improvement in Test Accuracy Relative to Random Grouping		
	optimal	(ours)	worst
a1	2.60%	2.6%	-3.03%
a2	1.53%	1.29%	-1.95%
a3	2.37%	1.72%	-3.04%
a4	2.67%	0.72%	-4.14%
a5	2.75%	2.29%	-3.87%
a6	2.04%	0.00%	-2.08%
a7	2.40%	0.74%	-2.43%
a8	1.61%	-1.59%	-1.59%
a9	8.38%	8.38%	-6.21%

Table 1: Performance of each task when trained with it's partner task *relative* to the expected performance from random groupings.



Multi-task Ablation Study

- Q2. How do inter-task affinities change over the course of training?
 - ✓ Affinity changes throughout the training, but no obvious pattern
 - ✓ However, the affinity rank remains
 - ✓ At initial stage of training, learned representation may extract general-purpose characteristics



4. Conclusion



Summary

- Contributions
 - I. Introduce systematic technique, TAG
 - II. Findings are supported by extensive analysis
 - III. Demonstrates improved performance

- Risks
 - I. Inter-task affinity is not inter-task similarity



Thank you

